

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #3: due Thursday, Feb. 5th, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

1. A student scavenges through a huge pile of old iPhones looking for one with a lot of memory. Suppose the probability distribution of phones in the pile is as follows. Let M be the amount of memory in a random phone the student encounters.

Phone	Probability	Memory (GB)
iPhone 3G	1/16	0.125
iPhone 4	11/64	0.5
iPhone 5	5/32	1
iPhone 6	5/32	1
iPhone 7	3/16	2
iPhone 8	1/8	3
iPhone 12	9/64	6

- (a) (5 points) Determine and sketch the PMF of M . Be careful with your drawing; we will be picky!
- (b) (10 points) Compute $E(M)$, using the direct formula for expected value. Show your work!
- (c) (10 points) Compute $Var(M)$, using the direct formula for expected value. Show your work!
2. The student can generally get more trade-in money for newer phones (defined as iPhone 6 or later) than for older phones. Suppose that a dealer will pay $5X$ dollars for each older phone, where X is a binomial random variable with $n = 6$ and $p = 0.3$, and $8Y$ dollars for each newer phone, where Y is a geometric random variable with $p = 0.4$. Let Z be the trade-in value of a random phone the student finds. Use the moments of random variables to answer the below questions (e.g., using Table 3.1, pp. 115–116 of Leon-Garcia or a similar table). This problem should not require a lot of computation.
- (a) (10 points) Compute $E(Z)$.
- (b) (20 points) Compute $E(Z^2)$.
3. Let the number of red phones the student finds be modeled as a Poisson random variable with an average of 1 red phone every 10 minutes.
- (a) (10 points.) What is the probability that the student finds no red phones within a given 15-minute exploration?
- (b) (10 points.) What is the probability that the student finds at least 3 red phones in 20 minutes of exploration?
4. The student grabs two independent items from the pile:
- C = the number of Lightning charging cables they find in a small side bin, modeled as a uniform random variable on the values $\{0, 1, 2, 3, 4\}$.
 - L = the number of Lightning-charging phones they find in a very short scan of the pile, modeled as a uniform random variable on the values $\{0, 1, 3, 8\}$.

Assume C and L are independent. The dealer offers the student a one-time bonus (in dollars) given by

$$N = 2C + 3L + 1.$$

- (a) (10 points) Compute $E(N)$.
- (b) (15 points) Compute $Var(N)$.